

Licensed Electrician's Theory (LET)

License Assessment

Sample Paper Two - RF (2025)



Candidate Surname	
Candidate Given Names	

Reference Material

- AS/NZS 3000:2018 Wiring Rules (including Amendments 1)
- AS/NZS 3012:2019 Electrical installations – Demolition and Construction sites
- AS/NZS 3008.1.1:2017 Electrical installations – Selection of cables
- Electricity Safety (General) Regulations 2019
- AS/NZS 4836:2011 Safe working on or near low voltage electrical installations and equipment



Instructions

- Personal notepads and paper are not permitted.
- Pens only must be used. Answers in pencil may not be marked.
- Do not remove any sheets from this assessment paper or the room.
- Papers with no name or signature will not be marked.
- Units must be shown to obtain full marks.

Results

Candidates need to obtain 75% or more to pass this assessment. If a mark of 74% or less is achieved, a minimum of 14 days is required before you are permitted to re attempt the assessment.

I, the above named candidate confirm:

- I understand the instructions provided to me.
- do not have any unauthorised materials in my possession
- have not attempted the Licensed Electrician's Theory Assessment at any venue within the past 14 days.

Candidate			
	Print name	Signature	Date

Working Time: 2 hours and 15 minutes

At the end of this time you will be asked to stop.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Total
Mark																			

Final Percentage	Pass/Fail

Supervisor			
	Print name	Signature	Date
Assessor			
	Print name	Signature	Date
Reviewed by (If necessary)			
	Print name	Signature	Date



AS/NZS 3000 WIRING RULES

In the following **four** Wiring Rules questions:

- You are required to write the Wiring Rules Clause and/or Table number in the space provided.
- The correct Wiring Rules Clause and Subclause must be given, e.g. 3.5.2(b)(i).

The correct answer to both parts must be given to obtain full marks.

Question 1. Is it a requirement that unattended submersible pump motors with a rating of 2.25kW be provided with over-temperature protection?

.....
Wiring Rules Clause Number: [2 + 2 = 4 Marks]

Question 2. Capacitors and electrical equipment containing capacitors shall be provided with a warning notice containing information similar to the following –

Warning: “.....
.....”

.....
Wiring Rules Clause Number: [2 + 2 = 4 Marks]

Question 3. AS/NZS3000:2018 Wiring Rules acknowledges two recognized hazardous areas. The two general recognized hazardous areas are:

- (a)
(b)

Wiring Rules Clause Number: [2 + 2 = 4 Marks]

Question 4. Where electrical equipment is connected to the fixed wiring by a connection in the form of a plug and socket-outlet, appliance plug or similar connecting device, any connection of exposed conductive parts to earth shall be:

.....
.....
.....
.....
.....
Wiring Rules Clause Number: [2 + 2 = 4 Marks]

AS/NZS 3012 CONSTRUCTION AND DEMOLITION SITES

In the following **two** AS/NZS 3012 questions:

- You are required to write the Standard's Clause number and/or Table number in the space provided.
- The correct Clause and Sub-clause number must be given. e.g. 2.10.2 (f).

The correct answer to both parts must be given to obtain full marks.

Question 5. Portable luminaries installed on construction or demolition sites shall be provided with a minimum degree of protection of

Standard Clause Number: [2 + 2 = 4 Marks]

Question 6. What requirements must be satisfied when supporting flexible cables on a catenary wire?

Standard Clause Number: [2 + 2 = 4 Marks]

ELECTRICITY SAFETY (GENERAL) REGULATIONS 2019

In the following Regulation question, you are required to:

- write your answers on the line/s below each question
- write the complete Regulation and Sub-Regulation number, if applicable, in the space provided, e.g. 401(e)(3).

The correct answer to both parts must be given to obtain full marks.

Question 7. The route of an underground electric line must be recorded as accurately as practicable within a margin of error not exceeding:

Regulation Number: [2 + 2 = 4 Marks]

ELECTRIC SHOCK SURVIVAL

Question 8. What ratio should Compression-to-ventilation be performed when performing CPR if a casualty is unresponsive and not breathing properly?

[2 + 2 = 4 Marks]

CABLE SELECTION**Question 9.**

THREE **four** core and earth V90 insulated and sheathed copper cables, are connected in parallel to supply a three-phase distribution board with a total maximum demand of 270A.

The cables are buried direct in the ground at a depth of 0.6m and are spaced 0.45m apart. They are protected by a circuit breaker.

- (i) Neglecting voltage drop, what is the minimum cable size which can be installed for this circuit.
- (ii) If the cables were spaced 0.15m apart instead of .45m, what is the minimum cable size which can be installed for this circuit?

All calculations including the final answer must be rounded to a maximum of **two decimal places**.

Table details and units must be shown below to obtain full marks.

Part (i)

	Answer		Answer		
Table 3 (_____?)		Item			
Table		Column			Answer
Derate/rating table		Column		Factor	
Derate/rating table		Column		Factor	

Part (i) Answer:

Question 9. Part (ii)

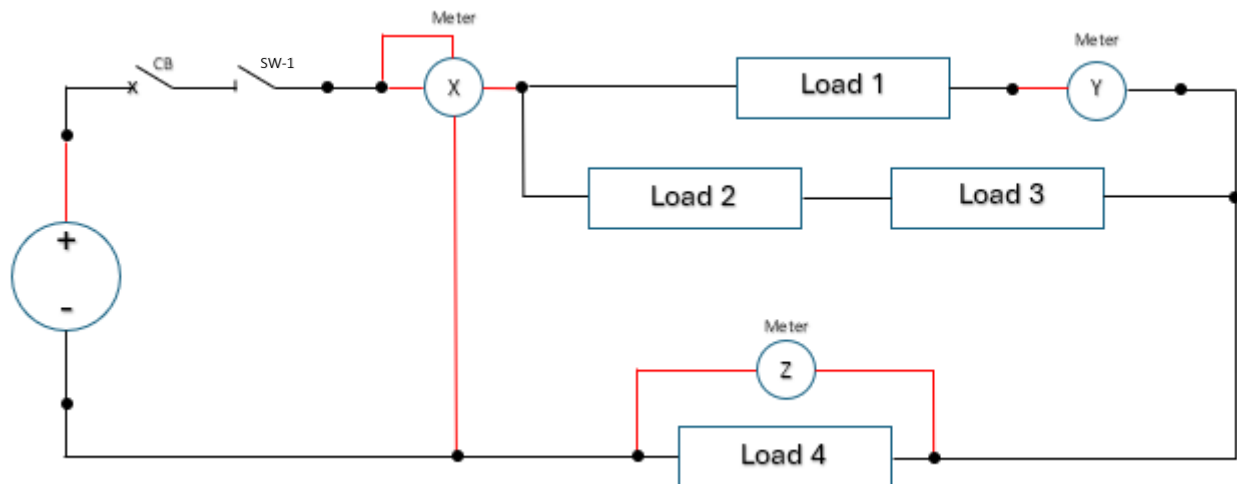
	Answer		Answer		Answer
Derate/rating table		Column		Factor	

Part (ii) Answer:

[1 + 2 + 1 + 1 + 2 + 1 = 8 Marks]

DC CIRCUITS

Question 10.



For the circuit shown above, you have taken measurements and recorded the following:

Supply Current – 800mA
 Voltage across load 1 – 57.6V
 Current through load 3 – 480mA
 Resistance of load 4 - 78Ω

Using these measurements, calculate the meter readings for the following when the switch and circuit breaker is closed.

- (i) Meter Y
- (ii) Meter X
- (iii) Meter Z

All calculations including the final answer must be rounded to a maximum of **two decimal places**.

Part (i) Answer :		Part (ii) Answer:		Part (iii) Answer:	
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[3 + 3 + 3 = 9 Marks]

MAXIMUM DEMAND**Question 11.**

Calculate the Maximum Demand for the mains of a single phase 230V domestic residence.

The load connected to the switchboard is:

- 1 – 9kW Range
- 1 – 2.8kW Spa pool heater
- 36 – 10A Double socket-outlets
- 9 – 10A Single socket-outlets
- 27 – (incandescent) Lighting points
- 1500W - Tennis court lighting
- 1 – 20A Socket-outlet

All calculations including the final answer must be rounded to a maximum of **two decimal places**.

All relevant table details, including table, column and load groups used.

Calculations and units must be shown to obtain full marks.

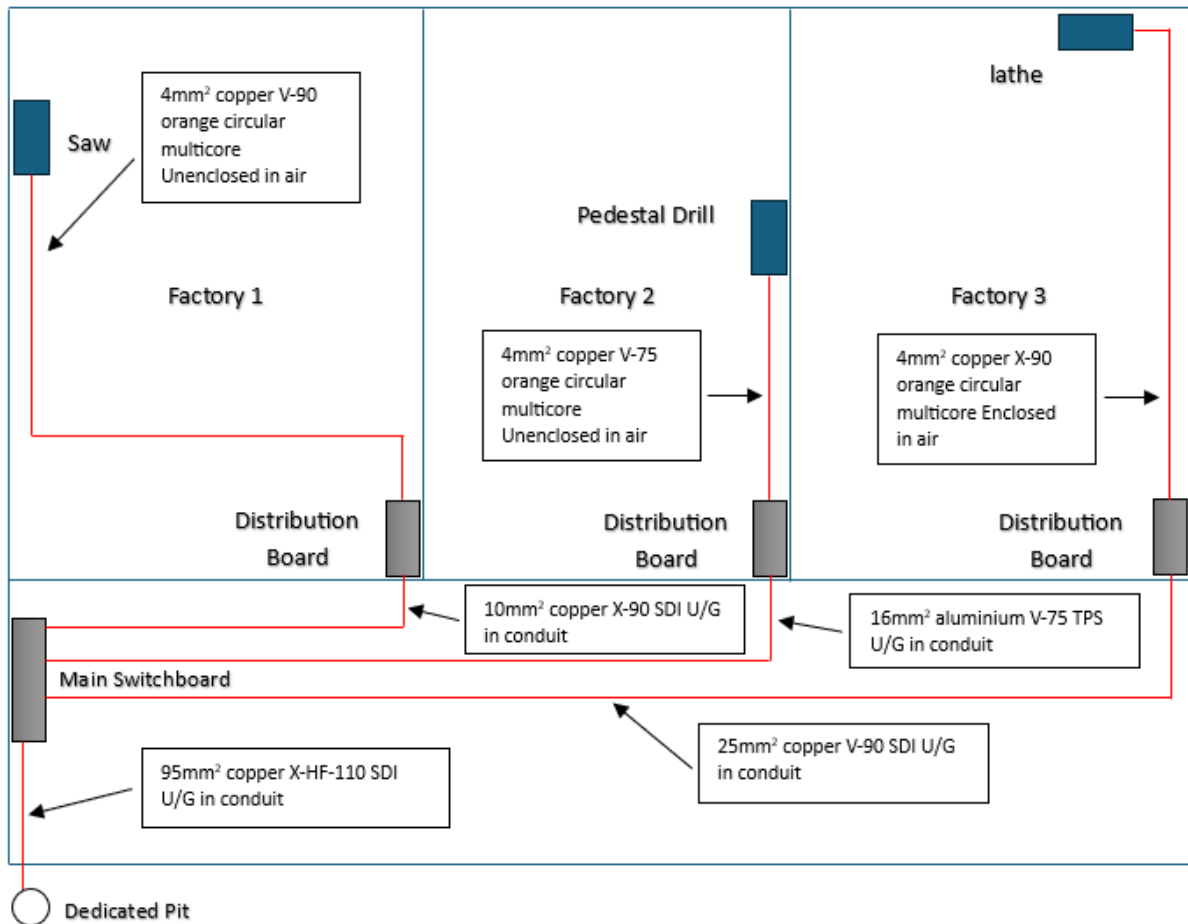
Table		Column	
Equipment	Load Group	Calculation	Maximum Demand
Total Maximum Demand:			

[1 + 1 + 1 + 1 + 1 + 2 + 1 = 8 Marks]

VOLTAGE DROP

Question 12.

In a 400/230V, three-phase commercial installation, a three-phase lathe is supplied from a sub-circuit originating at the distribution board in factory three. The lathe is installed 43 meters from the distribution board with a maximum demand of 28A.



Where Avenue

Client: Mike Hunt	Location	Cable	Distance	Maximum Demand	Voltage
Drawing title: Site plan	Main Switchboard	Consumers Mains	8m	120A	400/230V
Date: 12 February 2025	Factory 1 Distribution board	Sub Mains	50m	40A	400/230V
Location: 2-5 Where Ave. North Haverbrook	Factory 2 Distribution board	Sub Mains	70m	50A	400/230V
	Factory 3 Distribution board	Sub Mains	110m	60A	400/230V

Calculate the total voltage drop from the point of supply to the lathe terminals.

All calculations including the final answer must be rounded to a maximum of **two decimal places**.

All relevant table details, calculations and units must be show to obtain full marks.

Cable	Table	Column	Vc	Calculation	Vd
Consumer Mains					
Sub-mains					
Final Sub-circuit					

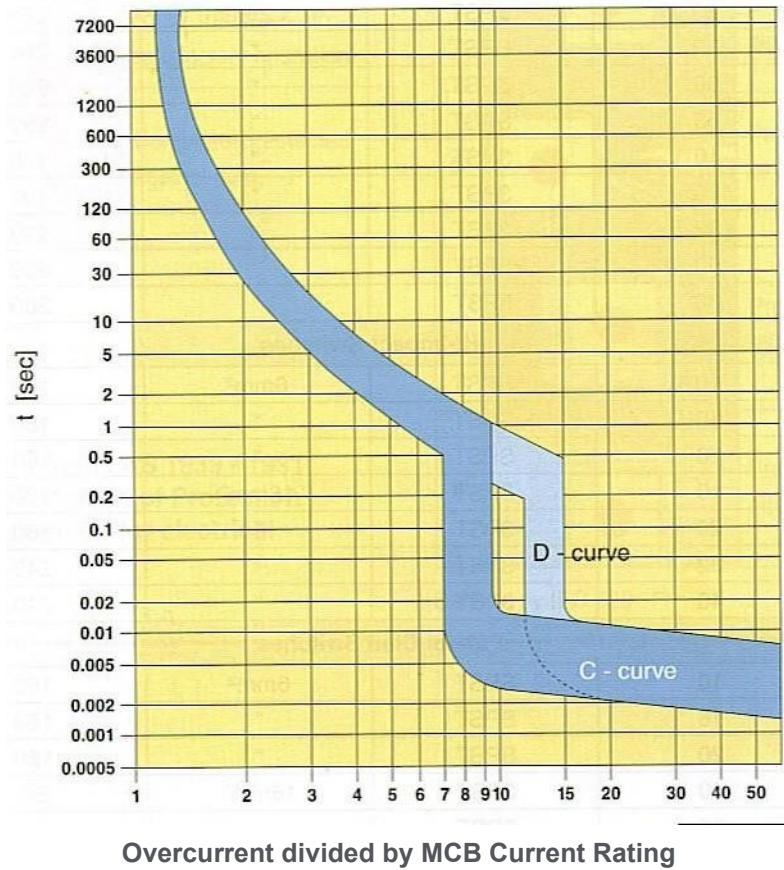
Answer Total Vd:	
------------------	--

[1+1+1+1+1+1+1+1+1+1 = 10 Marks]

OVERLOAD AND SHORT CIRCUIT CALCULATIONS

Question 13.

What are the minimum and maximum tripping times for a 20A Type C miniature over-current circuit breaker which is subjected to an over-current of 100A?



Overcurrent divided by MCB current rating:	
Minimum time:	Maximum time:

[1 + 1 + 1 = 3 Marks]

OVERLOAD AND SHORT CIRCUIT CALCULATIONS

Question 14.

The main switchboard of a 400/230V industrial installation is directly supplied from a 600kVA distribution transformer which has a prospective fault current of 18.6kA per phase. Sub-mains supply a distribution board from the main switchboard. Submains supply a distribution board from the main switchboard.

The following information is known:

- Impedance of the Consumers Mains = 0.00525 Ohms
- Impedance of the Sub-mains cables = 0.03480 Ohms

Determine the prospective fault current at:

- (i) the main switchboard; and
- (ii) the distribution board.

Work impedances to 5 decimal places.

All calculations must be shown to obtain full marks.

Transformer Impedance:	
MainSw/Bd:	
Dist/Bd:	

[(2+1) + (2+1) + (2+1) = 9 Marks]

RESIDUAL CURRENT DEVICES**Question 15.**

A 30mA Residual Current Device is to be installed supplying a distribution board. The maximum demand of the three circuits is 35A. There are three sub-circuits supplied from the distribution board, consisting of an 8A, 16A and 20A circuit breakers.

State the minimum current rating of the Residual Current Device.

Wiring Rules Clause Number: _____ [1 + 2 = 3 Marks]

MOTORS AND STARTERS**Question 16.**

CIRCLE the letter in front of the statement you consider to be the most correct.

On start up, a three-phase 400V squirrel cage induction motor with an automatic star/delta starter:

- A. starts on half the line voltage (200V)
- B. changes to delta connection at about 50% of the motors full rated speed
- C. has a starting current one half the DOL current
- D. has a starting torque one third of the DOL starting torque

[2 Marks]

AS/NZS 4836:2011**Question 17.**

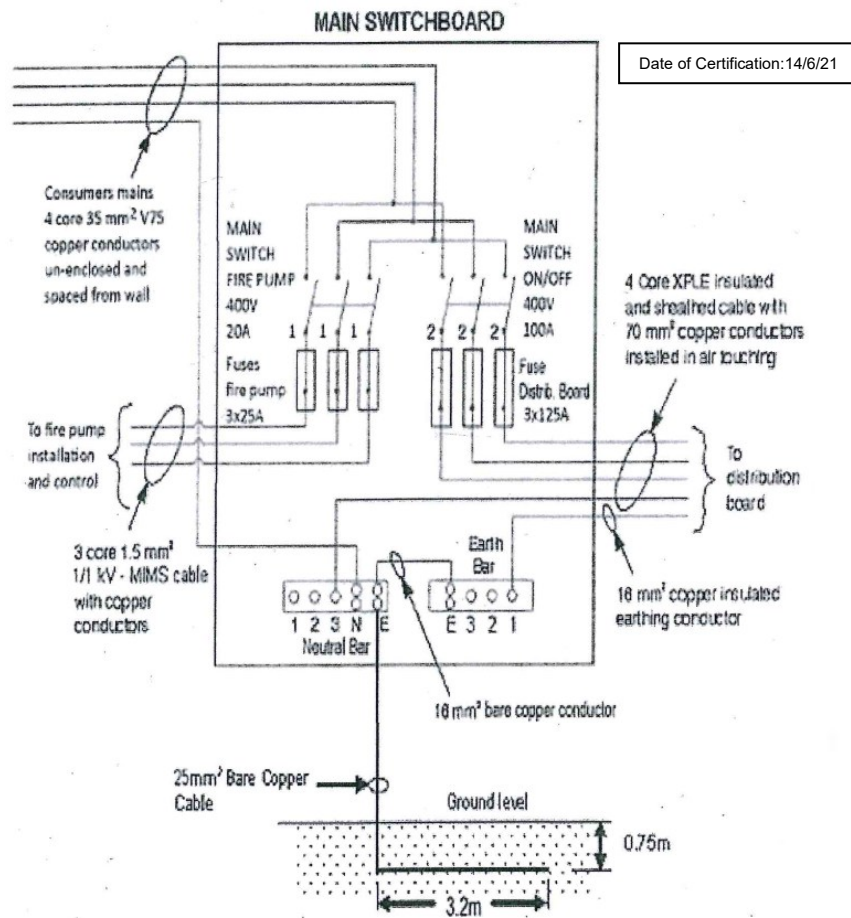
This question relates to AS/NZS 4836:2011.

Electrical equipment insulating covers rated for voltages less than 650 V shall comply and be tested to the requirements of which Standard?

Standard Clause Number: _____ [2 + 2 = 4 Marks]

INSTALLATION DEFECTS - NON DOMESTIC

Question 18.



The drawing above shows the MAIN SWITCHBOARD of an industrial installation originating at the consumers mains.

It supplies a distribution board having a connected load with a maximum demand of 110A per phase and an automatically controlled 3 phase fire pump motor having a current rating of 20A per phase. The fire pump does not require an RCD.

The multi-core MIMS cables are installed spaced from the wall.

Assume the MIMS cables are earthed in accordance with the Wiring Rules and are capable of maintaining supply to the equipment even when exposed to fire and mechanical damage.

All fuses are HRC type. The drawing contains contraventions to the Wiring Rules.

Emergency / Safety equipment installed in the Switchboard is separated from other Switchgear by a metal partition.

The Supply Authority has provided Short Circuit Protection of the point of Supply on the Consumers mains.

Complete the table on the following page.

Question 18. Continued.

Use the diagram on the previous page.

List **FOUR different defects** together with the contravened Wiring Rules Clause/Table number in the table provided below.

Note: Only the first four defects will be considered.

DEFECT DETAILS	WIRING RULE CLAUSE/TABLE No.

[4 x (2 + 1) = 12 Marks]